

NYCU Introduction to Machine Learning, Homework 4

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Part. 1, Kaggle (70% [50% comes from the competition]):

(20%) Introduction of your idea, methods, and key to success

Create a 3-5 page slide (e.g., MS PowerPoint, Google Slides, etc.) with no title or thank-you page to introduce your work.

Things you should cover - Include and not limit to

- How do you process the data? Have you done any special processing that significantly boosted performance?
- What is your model architecture? Do you make any changes/modifications to the model? Does it improve performance?
- How do you train the model? Do you use any special techniques (e.g., ensembles or other methods) to improve performance?
- Other details you want to mention that improve the performance
- Paste the snapshot of your Kaggle public leaderboard as an appendix (Not count to the page limit)

Hint:

1. Make your slide presentation clear and informative, and TAs will evaluate its completeness and content.
2. Read some literature to see how they report their method and results.

Part. 2, Questions (30%):

1. (10%) Based on the “SVMs vs. Logistic regression” lecture slide, explain which kinds of training data points mainly determine the classifier learned by SVM and which types of points influence Logistic Regression, and briefly justify your answer by referring to the shapes of hinge loss and logistic loss.

Type your answer here.

SVMs mainly determine by the data points that close to margin, which means that it focus on those which have “large possible to be made wrong prediction”. For those data points which reach “correct area”, they won’t provide contribution on updatation. See the blue curve (hinge loss), $z > 1$ part.

Compared to Logistic regression, red curve (logistic loss), although correct prediction would make those data point contribution decrease, close to 0, but they won’t become 0. And due to the $\ln$ function, outliers may have large influence (Left part, $z < 0$).

SVMs vs. Logistic regression

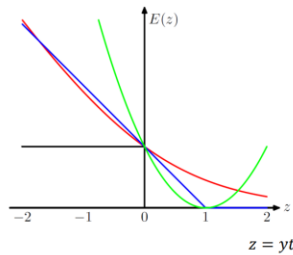
- Black curve
➤ misclassification error

- Red curve
➤ Logistic regression error

$$E_{LR}(yt) = \ln(1 + \exp(-yt))$$

- Blue curve
➤ Hinge error

$$E_{SV}(y_n t_n) = [1 - y_n t_n]_+$$



2. (15%) For an SVM with a Linear Kernel, determine whether to use the Primal or Dual Form for the datasets below. Justify your choice based on: (a) Optimization variables & computational complexity, (b) Memory requirements (specifically the size of the Gram Matrix), (c) Prediction cost.

- Dataset A: $N=100, M=20,000$
- Dataset B: $N=1,000,000, M=20$

Type your answer here.

Primal: ➤ Computational complexity is $O(M^3)$, where M is the number of optimization variables (number of data dimensionality)

Dual: It can be solved by using quadratic programming with complexity $O(N^3)$, where N is the number of training data

Testing phase

- Decision function of SVMs

$$y(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b$$

- Dual representation of SVMs

$$\mathbf{w} = \sum_{n=1}^N a_n t_n \phi(\mathbf{x}_n)$$

- For a test point $\mathbf{x}$, we classify it using a kernel function

$$y(\mathbf{x}) = \sum_{n=1}^N a_n t_n k(\mathbf{x}, \mathbf{x}_n) + b$$

- How to determine the value of b ?

For Dataset A, “Dual form” is better.

(a) Number of variable: Dual = $N = 100 <$ Primal = $M = 2e4$,

Dual computational complexity = $1e6 <$ Primal computational complexity $8e12$

(b) If dual, Gram Matrix size = $N*N = 1e5$

(c) Prediction cost: Dual $O(N*M) = 2e6$, Primal $O(M) = 2e4$

Computational complexity + Prediction cost: Dual $<$ Primal

For Dataset B, “Primal form” is better. Dual $<$ Primal

(a) Number of variable: Primal = $M = 20 <$ Dual = $N = 1e6$

Computational complexity: Primal = $8e3 <$ Dual = $1e18$

(b) If dual, Gram Matrix $N*N = 1e12$

(c) Prediction cost: Primal = $20 <$ Dual = $2e7$

Computational complexity + Prediction cost: Primal $<$ Dual

3. (5%) To train a neural network, what do we need to optimize it? (How do we know the net work is good or not?) Also, what algorithm can we use to optimize the neural network? (the most basic one).

Type your answer here.

Loss Function; Gradient Descent with Backpropagation.

By training loss and validation loss, observe the relation to judge if overfit (train low, val high) or learning rate too large (oscillation), or observe F1-score, ROC curve, confusion matrix etc. With these tools, we can understand more about the current situation of our model.